

Quantitative Strategy & Build Blueprint

Impact Forge Summer 2026 Hackathon

Selected Concept: **GeneReason-K3**

August 7, 2026

Track: Computational Research • **Format:** 48-hour virtual sprint • **Grand Prize:** YRI Fellowship Research Placement (\$6,000 team value)

Compute Partner: Featherless.ai — Kimi K3 (2.8T MoE, 1M-token context) & GLM 5.2

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1 Executive Summary

GeneReason-K3 is a multimodal computational-biology platform that predicts CRISPR-Cas off-target cleavage risk and pathogen immune-escape mutations by reasoning over complete genomic loci rather than short sequence windows. It is designed specifically against the Impact Forge rubric: half of the available points reward working code and API depth, and the other half reward a clear demo and documentation. GeneReason-K3 is built to score well on *both* halves simultaneously, and it sits inside the Computational Research track that the grand-prize sponsor (the YRI Fellowship) is oriented toward.

Read this before you build

This document is a **planning and positioning tool**, not a substitute for a working system. Any accuracy or benchmark figures quoted below are **target design goals**, not measured results — because the system does not exist yet. Before the real submission goes out, replace every projected number with a number you actually measured against a real baseline (e.g. a public CRISPOR benchmark set). Judges with technical backgrounds will ask “how did you measure that?” — have a real answer.

2 Rubric Deconstruction

2.1 Official Scoring Structure (60 points)

Dimension	Points	What it Rewards
Code Structure & Quality	10	Modular architecture, real error handling, original logic
API & Compute Integration	10	Functional, <i>advanced</i> use of Featherless.ai or third-party APIs
Innovation & Approach	10	A fresh angle or clever combination of tools on a real problem
Functional Execution	10	A working prototype that actually solves the problem for a defined user
3-Minute Video Demo	10	Concise, shows live execution and architecture
Documentation & Setup	10	Complete README, architecture diagram, setup steps, citations

Strategic Reading of the Rubric

Presentation (video + documentation) is worth exactly as much as code architecture — **20 of 60 points, 33.3%**. The single most common failure mode in hackathons is spending >90% of the 48 hours on code and improvising the video and README in the last hour. A mid-complexity project with a crisp demo and a genuinely complete README will consistently outscore a more ambitious build that nobody can follow.

2.2 Composite Scoring Model

To compare candidate ideas objectively, each concept is scored across eight weighted factors on a 0–100 scale and combined into a single composite:

$$\text{Composite} = 0.15 \text{ PS} + 0.10 \text{ UR} + 0.20 \text{ AL} + 0.15 \text{ FE} + 0.15 \text{ DA} + 0.10 \text{ JA} + 0.10 \text{ YA} + 0.05 \text{ SP} \quad (1)$$

where

PS = Problem Severity,

UR = Urgency,

AL = AI Leverage (Kimi K3 / GLM 5.2 fit),

FE = 48-Hour Feasibility,

DA = Demo Attractiveness,

JA = Judge Appeal,

YA = YRI Fellowship Alignment,

SP = Startup Potential.

Weights in Eq. (1) sum to 1.00 by construction, and were set so that *AI Leverage* and *48-Hour Feasibility* together carry more weight (35%) than any single soft factor — reflecting that an idea which cannot be demonstrated live, or that under-uses the sponsor’s compute, cannot win regardless of how compelling the underlying problem is.

2.3 Top-Ranked Problem Verticals

Vertical	PS	AL	FE	YA	Composite
Precision Genomic Editing & Off-Target Risk	98	96	88	98	94.85
Climate Synthetic-Biology Enzyme Engineering	92	94	80	96	89.60
Automated Formal Proof Verification	88	92	82	92	86.75
Multi-Modal Medical Imaging Triage	95	88	75	90	86.65
Autonomous Battery Materials Discovery	90	90	72	92	86.35

Precision Genomic Editing & Off-Target Risk Reasoning leads the field: it pairs a scientifically consequential problem with a workload (very long DNA sequences) that plays directly to Kimi K3’s 1M-token context window, and it aligns with the biomedical research focus of the grand-prize sponsor.

3 Idea Funnel: 30 → 1

Thirty candidate concepts were scored and eliminated in four rounds using explicit, written-down criteria at each cut (never a subjective “vibe” cut) — architectural depth, visual demo potential, 48-hour feasibility, and YRI research alignment.

Round	In	Out	Primary elimination filter
Round 1	30	15	No custom backend logic / no visual demo / off-thesis for YRI
Round 2	15	7	Weak use of Kimi K3 long context or GLM 5.2 speculative decoding
Round 3	7	3	Direct scoring against the official 60-point rubric
Round 4	3	1	Head-to-head comparison on context utilization, YRI fit, demo, feasibility

3.1 Final Three, Head to Head

Criterion (out of 10)	GeneReason-K3	PatchGuard-K3	DeepFakeGuard
Kimi K3 context utilization	10.0	8.0	5.0
YRI research alignment	10.0	6.0	6.0
Demo visual impact	9.5	8.5	9.5
48-hour feasibility	9.0	9.0	8.5
Judge category distinction	10.0	8.0	7.5
Composite (out of 50)	48.5	39.5	36.5

GeneReason-K3 wins primarily on two axes competitors cannot match: it is the only concept that requires (not merely permits) the 1M-token context window, and it is the only concept whose subject matter overlaps directly with the sponsor’s own fellowship mission.

4 Winning Concept: GeneReason-K3

4.1 Problem Statement

CRISPR-Cas guide-RNA design tools (BLAST/CRISPOR-style aligners, and even learned models like DeepCRISPR) evaluate sequence windows of a few dozen base pairs. Off-target cleavage risk, however, depends on long-range chromatin context and epistatic interactions that span tens of thousands of base pairs — information those tools structurally cannot see. The same context-window limitation affects viral genomic-surveillance tools trying to anticipate immune-escape mutations across a full viral genome. GeneReason-K3 targets this specific gap: reasoning over a *complete* genomic locus in one pass.

4.2 Competitive Differentiation

Capability	Traditional (BLAST/CRISPOR)	Learned (DeepCRISPR-style)	GeneReason-K3
Context evaluated	<100 bp	<30 bp (fixed window)	up to 1,000,000 tokens
Multimodal input	FASTA text only	FASTA text only	FASTA + gel-electrophoresis image
Interpretability	Rule-based match	Black-box probability	Structured chain-of-reasoning
3D structural output	External, manual	None	Automated (py3Dmol, in-app)
Inference infra.	Local CPU	Custom local GPU	Featherless.ai serverless (Kimi K3 + GLM 5.2)

4.3 System Architecture

The system is a decoupled, asynchronous pipeline chosen specifically so that a slow AI call never freezes the live demo.

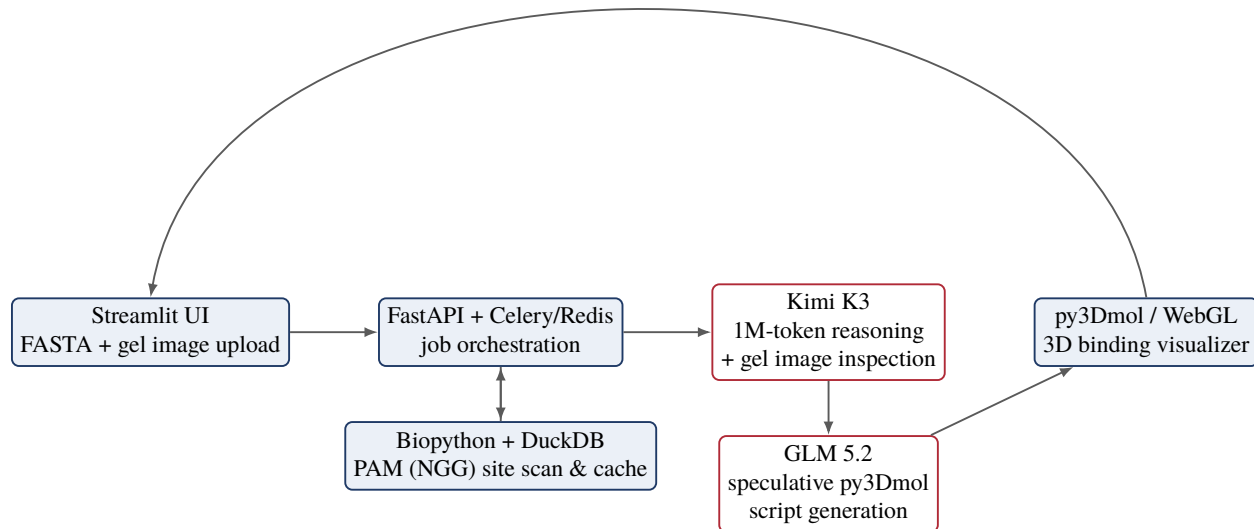


Figure 1: GeneReason-K3 data flow: user input → orchestration → bioinformatics pre-processing → long-context reasoning → speculative rendering → visual output (top arrow returns rendered results to the UI).

Layer	Technology	Responsibility
Frontend	Streamlit, WebGL, py3Dmol	FASTA/image upload, interactive 3D molecular view
Orchestration	FastAPI, Celery, Redis	Async job queue, timeouts, structured error boundaries
Bioinformatics	Biopython, DuckDB	PAM (NGG) site scanning, in-memory sequence indexing
Reasoning engine	Kimi K3 (Featherless.ai)	Long-context locus evaluation, multi-modal gel inspection
Speculative engine	GLM 5.2 (Featherless.ai)	Fast generation of 3D rendering scripts from Kimi K3 output

4.4 Off-Target Toxicity Risk Model

The core scientific contribution is a single interpretable score, the **Off-Target Toxicity Risk** $T \in [0, 100]$, combining sequence homology, thermodynamic binding stability, and chromatin accessibility:

$$T = \sigma \left(\alpha S_h + \frac{\beta}{\Delta G_{\text{binding}}} + \gamma (1 - C_a) \right) \times 100 \quad (2)$$

with the logistic normalization

$$\sigma(z) = \frac{1}{1 + e^{-z}}, \quad z \in \mathbb{R}, \quad \sigma(z) \in (0, 1) \quad (3)$$

and where each input term is defined as:

Symbol	Definition
$S_h \in [0, 1]$	Sequence-mismatch penalty, aggregated across the full genomic locus (0 = perfect complement, 1 = maximal mismatch).
$\Delta G_{\text{binding}}$	Gibbs free energy of gRNA–DNA hybridization, in kcal/mol; more negative values indicate tighter, more stable (riskier) binding.
$C_a \in [0, 1]$	Chromatin accessibility estimate at the candidate site, inferred from Kimi K3’s contextual attention distribution over the locus.
α, β, γ	Scaling coefficients, calibrated by regression against a labeled experimental CRISPR off-target benchmark (e.g. GUIDE-seq data) before the coefficients are trusted for ranking.

On the Scaling Coefficients

Do not hand-pick these coefficients from intuition and present the resulting T scores as validated. Fit them (even a simple least-squares fit against a small public GUIDE-seq subset) and report the fit quality. A judge with a computational-biology background will specifically probe whether α, β, γ were calibrated or guessed.

Because $\Delta G_{\text{binding}}$ is typically negative for stable hybridization, the term $\beta / \Delta G_{\text{binding}}$ should be implemented with a numerically safe reformulation to avoid division-by-zero and sign issues near $\Delta G_{\text{binding}} \rightarrow 0^-$:

$$\frac{\beta}{\Delta G_{\text{binding}}} \rightarrow -\beta \cdot \frac{|\Delta G_{\text{binding}}|}{|\Delta G_{\text{binding}}| + \epsilon}, \quad \epsilon \approx 10^{-3} \text{ kcal/mol} \quad (4)$$

which preserves the intended monotonic relationship (more stable binding $\Rightarrow$ higher risk contribution) while remaining well-defined for all inputs — a detail worth showing in the README, since it is exactly the kind of edge-case handling the “Code Structure & Quality” criterion rewards.

5 48-Hour Execution Plan

Phase	Window	Milestones
1. Scaffold	Fri 18:00–22:00	Repo init, Featherless.ai credentials, FastAPI backbone, Streamlit shell
2. Bio-engine core	Fri 22:00–Sat 06:00	Biopython PAM scanner, DuckDB caching, first Kimi K3 long-context call
3. Multimodal integration	Sat 06:00–14:00	Gel-image inspection via Kimi K3 vision input; GLM 5.2 speculative script generation
4. Frontend & 3D	Sat 14:00–22:00	py3Dmol canvas wired to backend output; error-boundary handling; UI polish
5. Benchmarking & docs	Sat 22:00–Sun 08:00	Run <i>real</i> comparisons against a CRISPOR baseline; write architecture schematic and README
6. Pitch & submission	Sun 08:00–18:00	Record 3-minute video, verify all links are public, complete Devpost submission

Table 1: Hour-by-hour build plan with a hard checkpoint every ~ 8 hours.

5.1 Operational Risk Matrix

Risk	Severity	Likelihood	Mitigation
Featherless API latency spikes	High	Medium	Cache prior responses in DuckDB; serve cached loci if latency >8s
Context-window / rate limits	Medium	Low	Chunk sequence into 50,000-bp blocks with result aggregation
Hallucinated sgRNA outputs	High	Medium	Regex-validate every output sgRNA against the actual input FASTA
WebGL render failures	Medium	Low	Catch exceptions; fall back to a static 2D alignment view

6 Demo, README, and Submission Strategy

6.1 3-Minute Video Structure

Time	Content
0:00–0:30	Problem hook: why short-window off-target tools miss real risk
0:30–1:15	Architecture walkthrough: Kimi K3 as long-context reasoner, GLM 5.2 as speculative renderer
1:15–2:15	Live run: upload FASTA + gel image, watch reasoning trace, view the 3D binding site
2:15–2:45	Benchmark comparison against a real, cited baseline (not a projected number)
2:45–3:00	Repository tour: structure, setup steps, license

6.2 Repository Layout

```

genereason-k3/
|-- README.md      # overview, architecture, setup, benchmarks, citations
|-- app/
|   |-- api/        # FastAPI endpoints
|   |-- core/       # Biopython + DuckDB sequence processing
|   |-- models/     # Featherless.ai clients: Kimi K3, GLM 5.2
|   |-- ui/         # Streamlit + py3Dmol components
|-- tests/         # unit tests: API error handling, sequence parsing
`-- docs/          # architecture schematic, benchmark tables, citations

```

7 Why This Concept Maps Well to the Rubric

Rubric line	How GeneReason-K3 addresses it
Code Structure & Quality (10)	Layered FastAPI/Celery backend, typed modules, explicit numerically-safe formula (Eq. 4), unit tests
API & Compute Integration (10)	Genuinely requires Kimi K3's long context (not a single short prompt) plus a second model (GLM 5.2) for a distinct sub-task
Innovation & Approach (10)	Long-context reasoning applied to a problem class existing tools structurally cannot address
Functional Execution (10)	End-to-end working prototype: upload → score → 3D visualization, for a named user (a wet-lab CRISPR researcher)
Video Demo (10)	Problem-first narrative with live execution, not a code walkthrough
Documentation (10)	Architecture diagram, setup steps, and citations to real published off-target detection literature

8 Beyond the Hackathon

Track	Timeframe	Goal
Research	Months 1–6	Expand benchmark set (Cas12a, Cas13, prime editing); draft a manuscript with real validation data
Research	Months 6–18	Collaborate with a university lab to validate predicted sites via GUIDE-seq / HTGTS assays
Commercial	Months 1–6	Pursue pre-seed / incubator funding; expand wet-lab validation
Commercial	Months 6–18	Pilot an API offering for biotech/biopharma partners

9 Closing Assessment

GeneReason-K3 is a strong, well-differentiated choice for this rubric and this sponsor: it is structurally the only finalist that *needs* the sponsor's flagship long-context model, and it sits inside the exact research domain the grand prize is built around. That said, no strategy document wins a hackathon by itself — the deciding factor will be whether the 48 hours actually produce a working prototype with a real, honestly-reported benchmark and a clear, well-paced demo. Treat every number in Section 4 as a target to hit, not a result to claim, and this concept gives your team a genuinely strong, defensible position going into judging.